17.2 Motion, velocity, and acceleration

Let an object have position described by

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$$

Then the velocity of the object is given by

$$\vec{v}(t) = \vec{r}\,'(t) = \langle f'(t), g'(t), h'(t) \rangle.$$

Facts: (1) For a given time t_0, $\vec{v}(t_0)$ is tangent to the trajectory of the object at $\vec{r}(t_0)$.

(2) The magnitude $\|\vec{v}(t_0)\|$ is the speed of the object at time t_0.

The acceleration of the object is

$$\vec{a}(t) = \vec{r}\,''(t) = \langle f''(t), g''(t), h''(t) \rangle.$$

→ Do all problems.